



Contents lists available at SciVerse ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

An experimental and kinetic modeling study of 2-methyltetrahydrofuran flames

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ARTICLE INFO

Article history:

Received 6 May 2013

Received in revised form 4 July 2013

Accepted 4 July 2013

Available online xxxx

Keywords:

2-Methyltetrahydrofuran

Laminar premixed flame

Molecular beam mass spectrometry

Electron ionization

Photo-ionization

Combustion model

ABSTRACT

Clean combustion processes are of paramount importance in the transition of the energy system towards increased sustainability. In an attempt to partially replace conventional fossil fuels, bio-derived oxygenates attract rising attention as alternative transportation fuels. Among this class of fuels, cyclic structures that can be derived from cellulosic biomass are particularly interesting. Here we present a study of premixed, laminar low-pressure flames of 2-methyltetrahydrofuran (2-MTHF) with an equivalence ratio of $\phi = 1.7$ at 40 mbar. Time-of-flight molecular-beam mass spectrometry (MBMS) with electron ionization (EI) was used to analyze and quantify mole fraction profiles of reactants, products, and most intermediate species including radicals involved in the combustion process. As a valuable complement, MBMS using single-photon ionization (PI) by vacuum ultraviolet radiation permitted isomer identification as well as independent concentration information under similar flame conditions. A detailed combustion model for 2-MTHF was developed, and the flame structure and species information were examined in conjunction with these experiments.

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1. Introduction

It is widely accepted that access to sustainable and clean energy is an urgent necessity. Combustion provides the major share of world energy today and will remain a substantial basis for the energy scenario in the near future. To reduce combustion-emitted carbon dioxide, biofuels including alcohols and esters have been promoted as interesting and promising alternatives to conventional petroleum-based fuels, and their combustion chemistry has already been studied in some detail [1]. Fuels that avoid the competition to food are especially valuable, however, and in this context, cyclic ethers including furan and its derivatives receive increasing attention because they are accessible from cellulosic biomass carbohydrates [2,3]. Among these cyclic ethers, 2-methyltetrahydrofuran (2-MTHF) is regarded as one of the particularly promising replacement fuels. Geilen et al. [4] have recently published a strategy to synthesize 2-MTHF from levulinic acid which is accessible from wood-based feedstocks via the molecular transformation to γ -valerolactone and 1,4-pentanediol. The pathway from cellulose to 2-MTHF and the potential of its energetically rea-

sonable large-scale production have been analyzed by Marquardt et al. [5]. The feasibility of using 2-MTHF in engines as a fuel extender was demonstrated already quite early in a spark-ignition engine [6]. More recently, Janssen et al. [7] have identified a blend of 2-MTHF with di-*n*-butylether as a fuel with optimal characteristics for diesel combustion and have investigated its use in a single-cylinder diesel engine. However, for a potential biofuel to be used in practice, its combustion chemistry should be known in more detail, especially regarding potentially harmful emissions. While several experiments and theoretical studies are available for furan [8–11] and the structurally related fuels 2-methylfuran (2-MF) [9,12,13], 2,3-dimethylfuran (2,3-DMF) [14,15], 2,5-dimethylfuran (2,5-DMF) [9,16–20] and tetrahydrofuran (THF) [15,21,22], detailed information on the combustion chemistry of 2-MTHF is scarce. Simmie et al. [23] have recently performed a theoretical study of the combustion kinetics of 2-MTHF, THF and 2,5-dimethyltetrahydrofuran (2,5-DMTHF). Mole fractions from premixed laminar 2-MTHF flames which are important for combustion model validation have, to the best of our knowledge, not been reported to date. For related systems, laminar premixed low-pressure furan flames were recently investigated by Tian et al. [24] using molecular-beam mass spectrometry (MBMS) with synchrotron-based tunable VUV photoionization (PI); they developed a kinetic model to

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interpret these data. More recently, a systematic study of the combustion chemistry of furan [11], MF [13], and 2,5-DMF [19] combustion under identical flame conditions was performed combining data from electron-ionization (EI) MBMS and gas chromatography (GC). These data were compared and analyzed with a single, somewhat updated kinetic model that was previously developed for 2,5-DMF [20] and contains the furan and MF sub-mechanisms. Kasper et al. [25] studied reaction pathways in premixed laminar flames of THF also using MBMS. Besides the identification of important combustion intermediates and their isomeric composition, they placed special emphasis on the resolution of the first reaction steps in the fuel destruction.

The present study, as the first detailed premixed flame structure investigation for 2-MTHF, focuses on the determination of quantitative species profiles in laminar, premixed, low-pressure, fuel-rich 2-MTHF flames by a combination of EI- and PI-MBMS. A detailed kinetic model has been developed for the high-temperature combustion of 2-MTHF, and important fuel destruction, oxidation, and growth reactions in this flame have been analyzed on the basis of the experimental data and the model.

2. Experiment

Fuel-rich laminar premixed 2-MTHF flames at $\phi = 1.7$ and 40 mbar were investigated with two independent MBMS instruments, with EI-MBMS in Bielefeld and PI-MBMS at the Advanced Light Source (ALS) in Berkeley. The high mass resolution of the EI-MBMS setup allowed distinguishing species by their elemental composition whereas the PI-MBMS set-up offered isomer separation by high energy resolution. The combination of both methods provided detailed information on reactive and stable intermediate species concentrations and thus facilitated a first validation of the kinetic mechanism for 2-MTHF combustion developed here. Although both experiments followed the same principles except for the ionization process, the two instruments are slightly different, and relevant details on set-ups and flame conditions will thus be given.

2.1. Electron-Ionization (EI) MBMS instrument

Experiments using electron ionization were carried out at Bielefeld University with a set-up described earlier in [26,27]. The reflectron time-of-flight (TOF) mass spectrometer is combined with a two-stage Wiley–McLaren ion source and enables a mass resolution of $m/\Delta m = 4000$. Thus, determination of the elemental composition of C/H/O species by their exact mass is possible in the relevant mass range. The burner is mounted in an upright fashion in the center of the flame chamber (320 mm diameter, 550 mm height). The flame gases were sampled via a quartz nozzle with a 25° cone angle and a $\sim 600 \mu\text{m}$ diameter orifice at the tip which was chosen to minimize perturbation effects [28]. A pressure reduction from 40 mbar in the flame chamber to $\sim 10^{-4}$ mbar in the 1st pumping stage interrupted further reactions and resulted in the formation of a molecular beam that passed through a copper skimmer (2 mm diameter) into the ionization chamber of the mass spectrometer at 10^{-6} mbar. Ionization relied on a pulsed electron beam of $\sim 10^9$ electrons/pulse with an energy distribution of 1 eV (Full Width at Half Maximum, FWHM) which permitted to detect all species in the same mass spectrum. Different soft ionization energies (10.6 eV, 12.0 eV, 13.4 eV, and 18 eV) were used to analyze the flame and to detect and minimize potential fragmentation reactions.

2.2. Photoionization (PI) MBMS instrument (Berkeley)

Single-photon ionization experiments were performed using synchrotron-generated vacuum ultraviolet (VUV) radiation at the

Advanced Light Source (ALS) in Berkeley. Details of the experimental set-up can be found in [29]. At the time of the present measurements, the MBMS system exhibited a mass resolution that was one order of magnitude lower than that of the Bielefeld set-up ($m/\Delta m = 400$). Its high energy-resolution of $\Delta E = \pm 0.05$ eV allows isomer identification of species by their ionization threshold (IP), as e.g. for allene (IP: 9.692 eV) and propyne (IP: 10.36 eV). The flame chamber (140 mm diameter) and the sampling system are arranged horizontally. A quartz nozzle with a cone angle of 40° and an orifice of $\sim 500 \mu\text{m}$ was used for sampling from the flame. The resulting molecular beam was then guided through a copper skimmer and crossed 260 mm behind the cone tip with the VUV beam of typically $\sim 10^{13}$ photons/s. Mass separation was performed using a linear Wiley–McLaren TOF mass spectrometer. Different soft ionization energies of 8.96 eV, 9.66 eV, 9.96 eV, 10.36 eV, 10.96 eV, 11.46 eV, 12.26 eV, 14.31 eV and 16.5 eV were used for the analysis of the flame.

2.3. Flame conditions

Fuel-rich ($\phi = 1.7$) laminar flat 2-MTHF/O₂/Ar flames were stabilized on McKenna-type burners at 40 mbar. The stoichiometry was chosen because a fuel-rich flame provides a large intermediate pool for analysis. Two different burners were used. The EI-MBMS instrument featured a home-made burner with a bronze sintered matrix of 64 mm diameter, while the PI-MBMS experiment was equipped with a commercial stainless-steel McKenna burner of 60 mm diameter. The conditions that were chosen to stabilize the flames in both systems were slightly different; they are listed in Table 1. The cold gas (300 K; 40 mbar) velocity was 32.8 cm/s for the EI-MBMS set-up and 28.2 cm/s for the PI-MBMS instrument which corresponds to an overall mass flow of 4.246 g/min and 3.233 g/min, respectively. The flow rates of oxygen and argon were regulated by calibrated mass-flow controller (MKS, Aera) with a precision of $\pm 5\%$. The dosage of the liquid 2-MTHF fuel was realized by a syringe pump injecting it into a heated vaporizer (100 °C). Burners were mounted on translational stages so that the distance from the quartz nozzle could be varied. By varying this distance, flame concentration profiles were obtained along the centerline of the flame as a function of height above the burner (h). The pressure was kept constant during the measurements using a downstream butterfly valve (MKS) regulating the pumping speed of a mechanical pump.

2.4. Data evaluation

The data quantification method was adapted from previous work [26,30]. A detailed description of the procedure used in the data evaluation of the EI-MBMS data is given in the supplement of [31]. Only the main aspects of the evaluation methods are thus reported here.

For both experiments the integrated ion signal S_i of a species i is correlated to its mole fraction by the following relationship:

$$S_i = x_i \cdot c \cdot \phi \cdot SW \cdot D(M_i) \cdot FKT \cdot \int \sigma_i(\tau) \cdot f(E - \tau) d\tau \quad (1)$$

Here, x_i is the mole fraction of species i , c an instrument factor, ϕ the number of electrons or photons, SW the number of sweeps, $D(M_i)$ the mass discrimination factor of species i . FKT is a

Table 1
Flame conditions, slm: standard liters per minute.

ϕ	C ₅ H ₁₀ O (slm)	O ₂ (slm)	Ar (slm)	Total (slm)	Velocity (cm/s)	MBMS setup
1.7	0.33	1.3767	0.5689	2.28	32.8	Bielefeld/ EI
1.7	0.29	1.21	0.22	1.72	28.2	ALS/PI

temperature- and thus position-depending sampling function, and $\int \sigma_i(\tau) \cdot f(E - \tau) d\tau$ is the convolution of the energy distribution of ionizing particles with the ionization cross section (see [30] for details), with τ as integration variable. Eq. (1) may be simplified if a species is measured relative to a reference species R :

$$\frac{S_i}{S_R} = \frac{x_i}{x_R} \cdot \frac{D(M_i)}{D(M_R)} \cdot \frac{\int \sigma_i(\tau) \cdot f(E - \tau) d\tau}{\int \sigma_R(\tau) \cdot f(E - \tau) d\tau} = \frac{x_i}{x_R} \cdot k_{i/R}(E) \quad (2)$$

Here, $k_{i/R}(E)$ is a calibration factor which combines all species-dependent parameters and which is related to the ionization energy. The EI-MBMS instrument, because of the broad energy distribution of the electron beam, enables a species overview from one single mass spectrum. Thus, argon was used as a reference signal at all measured energies. The calibration factors $k_{i/Ar}(E)$ were generated from cold gas measurements of known gas compositions (*direct calibration*), or were estimated by the RICS (*relative ionization cross sections*) method [32], or used a convolution of literature ionization cross sections with the known kinetic energy distribution of ionizing electrons [30]. With a known calibration factor, the mole fraction of each species was obtained based on the argon mole fraction profile from the main species calculation (see below). Additional signal corrections were applied to correct for ^{13}C and ^{18}O isomers and background signals. To reduce the influence of fragmentation reactions on mole fraction profiles, signals were corrected for fragmentation coming from species with known fragmentation patterns. Furthermore, the scan with the lowest possible energy was used for most of the species.

In the PI-MBMS experiment, the argon signal cannot be measured below its ionization threshold of 15.759 eV and thus cannot be used as a reference. The evaluation of species mole fractions therefore resulted from Eq. (1) and the direct use of tabulated photoionization cross sections. For this, the mass discrimination factors, the sampling function, and the photon flux were determined individually. Mass discrimination factors were determined experimentally and the photon flux was documented during the measurements with a calibrated photodiode. To determine the sampling function FKT and the instrument factor c , Eq. (1) was solved for

$$c \cdot FKT = \frac{S_i}{x_i \cdot \Phi \cdot SW \cdot D(M_i) \cdot \int \sigma_i(\tau) \cdot f(E - \tau) d\tau} \quad (3)$$

This product was obtained from the argon signal (measured at 16.61 eV) and its mole fraction profile adapted from the major species evaluation. For this, it was assumed that the instrument factor c was identical for all scans. Raw signal corrections were also performed for ^{13}C and ^{18}O isomers. Because of the high energy resolution of the PI-MBMS experiment, fragmentation can be excluded in the majority of cases when evaluating the scan with the lowest available photon energy above the ionization threshold of the species. If more than one species was measured at a nominal mass, Eq. (1) was used in reverse to calculate and subtract signal contributions from species with lower ionization thresholds.

Mole fractions of major species including 2-MTHF, O_2 , H_2 , H_2O , CO , CO_2 , and Ar were calculated for both instruments in an analogous manner using Eq. (2). All signals were taken from the 18 eV scan for the EI-MBMS instrument, and for the PI-MBMS instrument, species signals were taken from the scan closest in energy to the ionization threshold, i.e. 10.96 eV for 2-MTHF, 12.26 eV for O_2 , 16.61 eV for H_2 , 13.16 eV for H_2O , 14.31 eV for both CO and CO_2 , and 16.61 eV for Ar. The determination of the necessary calibration factors was performed using an internal calibration strategy [30] based on the elemental C, H, and O balances and on a single cold-gas calibration measurement of a defined CO/CO_2 mixture. The argon mole fraction was calculated as the balance between the sum of all major species and unity [31]. To account for

early consumption of small amounts of O_2 and the fuel near the burner surface, the individual element balances were used to determine calibration factors for these species. Furthermore, a correction of the raw signals for their background (caused by accumulation of background gas in the ionization chamber) was done for H_2 , CO , CO_2 , and H_2O .

Although many corrections have been carefully considered, mole fractions may be affected by several uncertainties. Even if care is taken to correct for fragmentation (especially from the fuel), smaller contributions for species with unknown fragmentation patterns, e.g. radicals, cannot be excluded. In the EI-MBMS experiment, uncertainties in the mole fractions depend on the accuracy of the calibration factors and on the respective electron ionization cross sections. The evaluation of mole fractions in the PI-MBMS experiment includes a large number of literature cross sections and measured signals. Therefore, the absolute mole fraction values tend to be less accurate than in the EI-MBMS experiment whereas the profile shapes are usually more reliable for PI-MBMS because of lesser importance of fragmentation reactions. The specific accuracy depends mainly on the quality of the respective photon ionization cross section. For both experiments uncertainties may be as low as 20%, depending on the detected species, but factors of 2–4 uncertainty are possible if cross sections must be estimated.

2.5. Temperature profiles

Temperature profiles were determined and used as input parameters for the kinetic modeling, recognizing that MBMS is an invasive technique and that the flow and temperature fields of the laminar flame are perturbed by the sampling probe. Typically the perturbation influence on the measured species profiles is lowest for sufficiently large burner-to-cone distances [28], and the influence on the temperature field can be described adequately by using a perturbed temperature profile [33,34]. Perturbed temperature profiles were obtained here from the temperature dependence of the sampling rate. The gas flow rate $\dot{N}$ through the sampling orifice can be described by the following relation:

$$\dot{N} \propto p_{1st} \propto \frac{S_{Ar}}{x_{Ar} \cdot \varphi} = C \cdot \sqrt{\frac{\gamma}{M \cdot T}} \cdot \left(\frac{2}{\gamma + 1} \right)^Z \quad (4)$$

with $Z = (\gamma + 1)/2(\gamma - 1)$.

Here, S_{Ar} is the integrated ion signal and x_{Ar} the mole fraction of argon, φ is the number of electrons, γ is the heat capacity coefficient (C_v/C_p), and M is the mean molar mass. C is an instrument-specific constant, which can be obtained at a defined position with known temperature T and composition. Such a position can be most reliably identified in the exhaust gas region where the temperature can be determined without the need of high spatial resolution and the gas composition is well known.

The temperature of the calibration point in the exhaust gas was measured by laser-induced fluorescence (LIF) of seeded NO at a distance of $h = 16$ mm without the presence of the sampling cone and for both flame conditions. Small amounts of NO (<0.5%) were added to the fuel stream for this measurement. NO was excited in the A–X (0–0) band in the range of 225.047–225.124 nm with a dye laser (Coherent, Scan Mate Pro) which was pumped by an excimer laser (Coherent, CompEx150) at 50 Hz. The broadband LIF emission was detected with a CCD camera (Andor, iStar DH734), and the temperature was evaluated with an accuracy of $\pm 5\%$ by direct fitting of the spectra.

3. Kinetic model development

A detailed chemical kinetic model has been constructed to capture and describe the high-temperature combustion of

2-methyl-tetrahydrofuran. The hierarchically constructed mechanism is presented and validated against these low-pressure flame data. In this hierarchical approach, the relevant species and classes of reactions have been included. The recent comprehensive mechanism of Burke et al. [35] was used for the H_2/O_2 chemistry. Details of the CO sub-mechanism have been described by Haas et al. [36], and their updates to these sub-mechanisms were included as such. A detailed mechanism for the combustion of THF was included to capture the reactions of THF and its primary radicals and combustion intermediates [37]. The latter contains C_1 – C_3 chemistry based on the work of Lowry et al. [38] and the C_4 chemistry of Healy et al. [39]. A subset to describe the formation of aromatic species such as benzene, fulvene, phenol, and toluene was added from Narayanaswamy et al. [40].

The kinetics of 2,5-DMTHF and the combustion of the related oxolanes THF and 2-MTHF have recently been studied theoretically by Simmie [23]. The rate coefficient expressions for abstraction of hydrogen atoms from 2,5-DMTHF by H and CH_3 were reported in that study along with the rate coefficient expressions for the subsequent ring-opening steps of the produced primary radicals. The corresponding rate expressions for 2-MTHF in the present mechanism are based upon these results. The other reaction rates for the 2-MTHF sub-mechanism including the relevant high-temperature reaction classes were estimated based on THF or other equivalent

species, as described below. Further study regarding the low-temperature combustion of this interesting next-generation biofuel is ongoing and a detailed mechanism describing both low- and high-temperature chemistry is currently being developed.

The thermodynamic and transport data for 2-MTHF and related radicals and combustion intermediates were estimated using the thermodynamic and transport data estimator of the Reaction Mechanism Generator software package of Green and coworkers [41]. More accurate detailed theoretical calculations of these data are under way and will be presented together with the low-temperature chemistry mechanism. For all other species included in the mechanism, the thermodynamic and transport data from the literature mechanisms mentioned above have been used.

The entire mechanism consists of 185 species and 1412 reactions, with 25 species and 168 reactions added for the 2-MTHF sub-mechanism. The complete mechanism is available as [Supplemental material](#) to this publication, with mechanism-, thermo- and transport-files in CHEMKIN syntax [42]. All simulations were carried out using the CHEMKIN-PRO package [42]. The flame simulations were conducted using the premixed laminar burner-stabilized flame reactor model to adaptive-grid-control-based solution gradient and curvature of 0.1, assuming mixture-averaged transport, accounting for thermal diffusion, and reaching a high resolution with approximately 300 grid points. Converging to stricter,

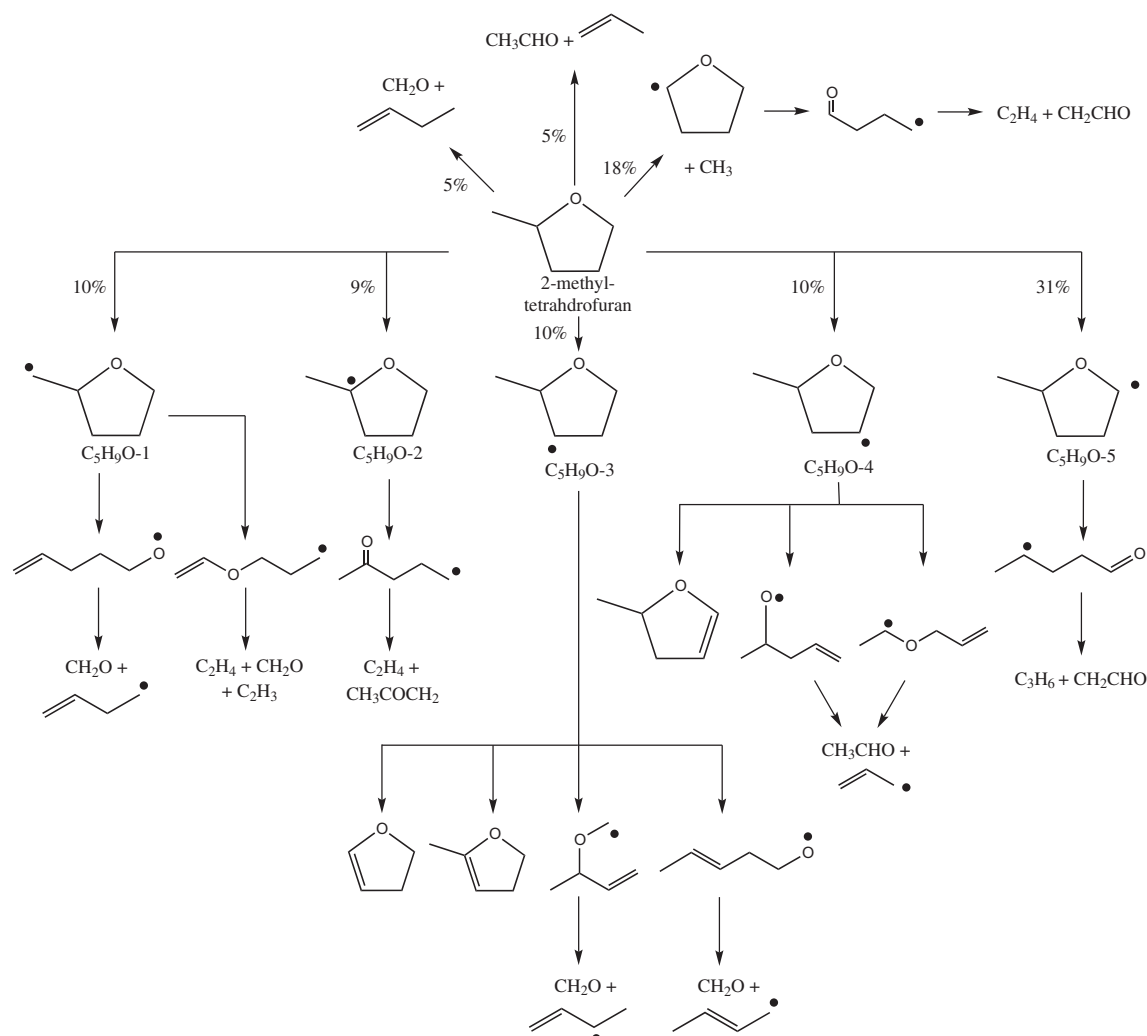


Fig. 1. Reaction pathways for the consumption of 2-methyltetrahydrofuran in a low-pressure, fuel-rich flame ($\phi = 1.7$). The percentages next to the arrows indicate the contribution of the respective pathway to fuel consumption.

lower GRAD and CURV settings did not significantly change any simulated species mole fraction profiles. For the simulations we have used temperature profiles in which the exhaust gas temperature was increased by the experimental uncertainty, 100 K, and the temperature profiles were adapted accordingly, with improved agreement with the experimental data (see also Section 4.1). The measured and adapted temperature profiles for both experiments are available in the [Supplemental material](#).

3.1. Fuel chemistry

Various reactions are possible for the oxidation of 2-MTHF and its fate depends on temperature and pressure conditions as well as on the reactants concentrations. The same applies for the different possible fuel radicals and combustion intermediates. Model simulated reaction pathways for the consumption of 2-MTHF are shown in [Fig. 1](#). The included reaction classes are presented here together with a discussion of corresponding rate constants.

3.2. Unimolecular fuel decomposition

Several decomposition reactions are possible for 2-MTHF. The scission of any of the ring hydrogen bonds in the molecule is listed in the reverse direction in the mechanism with the rate constant for the recombination of a hydrogen atom with one of the primary radicals estimated as $1 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ in accordance with Allara and Shaw [43]. Scission of the alkyl C–H bond in 2-MTHF is also listed in the reverse direction with the rate constant based on the recombination of a primary alkyl radical and a hydrogen atom. The most influential decomposition reaction is, however, the C–C bond split yielding CH_3 and a THF radical with the radical center on position two, as the methyl-ring bond is the weakest bond in the molecule. The rate for this reaction has been estimated by Simmie [23] from the methyl + ethyl recombination.

The elimination of stable molecules from the fuel is included as well, with the formation of formaldehyde and butene, acetaldehyde and propene or acetone and ethylene. The corresponding rate coefficient expressions are based on the equivalent elimination reactions for THF [37].

3.3. H-atom abstraction reactions

The rate constants for hydrogen abstraction by H and CH_3 have been theoretically calculated by Simmie for 2,5-DMTHF [23]. For 2-MTHF we have used these rates for the methyl site and abstraction from C2, C3 and C4, with all rates divided by two as 2,5-DMTHF has two equivalent sites for each radicals whereas 2-MTHF only has one. Abstraction from C5 by H and CH_3 is estimated to be equal to the rate for the abstraction from THF [37], again divided by two. No theoretical or experimental results are available for H-atom abstraction from 2-MTHF or 2,5-DMTHF by OH and HO_2 , these rates are therefore based on the THF chemistry. For abstraction from C5, C4, and C3, the THF rate expressions are applied, reduced by a factor two to account for the difference in symmetry. For the C2 and the methyl site, no equivalent site is present in THF, the rates are assumed to be the same as the estimate for the C3 site, as similar behavior is observed for the abstractions by H and CH_3 . These rough estimations will be improved further in the future through detailed theoretical calculations but prove less influential for the presented 2-MTHF flame study.

3.4. Fuel radical decomposition reactions

In his study on the kinetics of oxolanes, Simmie included the ring-opening reactions of the primary radicals [23]. The results from this accurate study are thus used to express the rate coefficients

Table 2

Arrhenius parameters for ring-opening reactions of primary fuel radicals and subsequent beta-scission reactions, rate coefficients based on Simmie [23].

Reaction	$A \text{ (s}^{-1}\text{)}$	n	$E_A \text{ (cal)}$
$\text{C}_5\text{H}_9\text{O-1} = \text{CH}_2\text{CH}_2\text{CH}_2\text{OC}_2\text{H}_5$	$5.0\text{E}+13$	0.0	21773
$\text{CH}_2\text{CH}_2\text{CH}_2\text{OC}_2\text{H}_5 = \text{C}_2\text{H}_4 + \text{C}_2\text{H}_5\text{OCH}_2$	$5.0\text{E}+13$	0.0	25358
$\text{C}_5\text{H}_9\text{O-1} = \text{OCH}_2\text{CH}_2\text{CH}_2\text{C}_2\text{H}_5$	$5.0\text{E}+13$	0.0	18355
$\text{OCH}_2\text{CH}_2\text{CH}_2\text{C}_2\text{H}_5 = \text{CH}_2\text{O} + \text{CH}_2\text{CH}_2\text{C}_2\text{H}_5$	$5.0\text{E}+13$	0.0	14029
$\text{C}_5\text{H}_9\text{O-2} = \text{CH}_2\text{CH}_2\text{OC}(\text{CH}_3)\text{CH}_3$	$7.4\text{E}+13$	0.0	32337
$\text{CH}_2\text{CH}_2\text{OC}(\text{CH}_3)\text{CH}_3 = \text{C}_2\text{H}_4 + \text{CH}_3\text{COCH}_2$	$6.9\text{E}+11$	0.0	14483
$\text{C}_5\text{H}_9\text{O-2} = \text{CH}_2\text{CH}_2\text{CH}_2\text{C}(\text{O})\text{CH}_3$	$4.9\text{E}+13$	0.0	21630
$\text{CH}_2\text{CH}_2\text{CH}_2\text{C}(\text{O})\text{CH}_3 = \text{C}_2\text{H}_4 + \text{CH}_3\text{COCH}_2$	$2.8\text{E}+12$	0.0	20674
$\text{C}_5\text{H}_9\text{O-3} = \text{CH}_2\text{OCH}(\text{CH}_3)\text{C}_2\text{H}_5$	$3.9\text{E}+13$	0.0	32337
$\text{CH}_2\text{OCH}(\text{CH}_3)\text{C}_2\text{H}_5 = \text{CH}_2\text{O} + \text{CH}_3\text{CHC}_2\text{H}_5$	$3.8\text{E}+13$	0.0	17471
$\text{C}_5\text{H}_9\text{O-3} = \text{OCH}_2\text{CH}_2\text{CHCHCH}_3$	$2.2\text{E}+13$	0.0	29588
$\text{OCH}_2\text{CH}_2\text{CHCHCH}_3 = \text{CH}_2\text{O} + \text{CH}_2\text{CHCHCH}_3$	$1.3\text{E}+13$	0.424	19048
$\text{C}_5\text{H}_9\text{O-4} = \text{CH}_2\text{CHCH}_2\text{CH}(\text{CH}_3)\text{O}$	$5.0\text{E}+13$	0.0	30951
$\text{CH}_2\text{CHCH}_2\text{CH}(\text{CH}_3)\text{O} = \text{CH}_2\text{CHCH}_2 + \text{CH}_3\text{CHO}$	$5.0\text{E}+13$	0.0	16491
$\text{C}_5\text{H}_9\text{O-4} = \text{CH}_2\text{CHCH}_2\text{OCHCH}_3$	$5.0\text{E}+13$	0.0	31620
$\text{CH}_2\text{CHCH}_2\text{OCHCH}_3 = \text{CH}_2\text{CHCH}_2 + \text{CH}_3\text{CHO}$	$5.0\text{E}+13$	0.0	15511
$\text{C}_5\text{H}_9\text{O-5} = \text{CH}_2\text{CHOCH}(\text{CH}_3)\text{CH}_2$	$5.0\text{E}+13$	0.0	33269
$\text{CH}_2\text{CHOCH}(\text{CH}_3)\text{CH}_2 = \text{C}_3\text{H}_6 + \text{CH}_2\text{CHO}$	$5.0\text{E}+13$	0.0	11042
$\text{C}_5\text{H}_9\text{O-5} = \text{CH}_3\text{CHCH}_2\text{CH}_2\text{CHO}$	$5.0\text{E}+13$	0.0	21821
$\text{CH}_3\text{CHCH}_2\text{CH}_2\text{CHO} = \text{C}_3\text{H}_6 + \text{CH}_2\text{CHO}$	$5.0\text{E}+13$	0.0	19789

as a function of temperature for the ring opening steps and the subsequent beta-scissions of the produced non-cyclic radicals. All reactions and rate constants are listed in [Table 2](#). Besides the ring-opening decompositions, also beta C–H bond scissions can take place yielding 2-methyl-dihydrofurans, of which three different isomers are possible. For these reactions the rate coefficient expressions can also be based on the work of Simmie for 2,5-DMTHF [23]. The exception is the C5 radical, the rate estimation for which is based on THF [37]. The beta C–H bond scission of the fuel radical with its radical center on the methyl C-atom ($\text{C}_5\text{H}_9\text{O-1}$) is too slow to have any influence due to the high barrier caused by the ring strain of an external double bond on the tetrahydrofuran ring.

3.5. Fuel isomerization

The radicals can isomerize by transferring H-atoms from other carbon atoms to the radical site and this reaction class is also captured in the mechanism. Due to the ring strain only isomerization between adjacent carbon atoms is possible. The rate expressions for a hydrogen atom transferring to the radical site from an adjacent carbon atom are estimated based on similar reactions [37].

3.6. Hydrogen atom abstractions from primary fuel radicals

The produced fuel radicals with the radical site on one of the ring carbon atoms can react with available radicals such as H, OH, O, O_2 , and CH_3 . These radicals can abstract a second H-atom from the ring leading to formation of one of the three 2-methyl-dihydrofuran isomers. The rate coefficients in the presented mechanism are estimated to be equal to the equivalent reactions of THF [37]. These reactions, especially the reaction with O_2 , can be significant production channels for 2-methyl-dihydrofuran and they can also lead to loss of a hydrogen atom from the fuel radicals through beta-scission.

3.7. Consumption of 2-methyl-dihydrofuran

An important class of stable intermediates in the combustion of 2-MTHF is 2-methyl-dihydrofuran ($\text{C}_5\text{H}_8\text{O}$). These unsaturated cyclic compounds can easily react via hydrogen abstraction from the

allylic site through one of the available radicals like H and OH. The produced resonance-stabilized radical can either decompose or recombine with an available hydrogen atom to yield 2-methyl-dihydrofuran again. Another possibility is the addition of OH to the double bond and subsequent decomposition through beta-scissions. Oxygen atoms can insert in the double bond leading to ring opening and further decomposition. All rate constants in this part of the mechanism are estimated from similar reactions of THF combustion intermediates and other ethers.

3.8. Other reactions

2,3-Dihydrofuran is an important combustion intermediate for 2-MTHF as it can be produced by CH₃ elimination from the primary radical with its radical site on C3 (C₅H₉O-3). In their shock-tube and quantum chemical studies of the pyrolysis and isomerization of 2,3-dihydrofuran, Lifshitz et al. [15] have presented the isomerization to cyclopropane carboxaldehyde, and further to 2-butenal (crotonaldehyde). These reactions are included with the rate constants proposed by Dubnikova and Lifshitz [14,15]. Both 2,3-dihydrofuran and 2-butenal have been experimentally identified in the 2-MTHF flame.

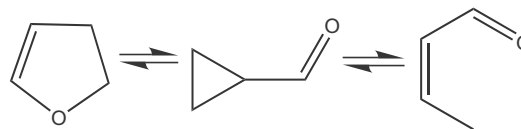


Table 3

Intermediate species measured with EI-MBMS in the 2-MTHF flame. *M*: nominal mass; *E*: electron energy; *IP*: ionization threshold; calibration method: RICS (of reference species in parenthesis), convolution (of energy distribution and ionization cross section) or direct (cold gas calibration); Ref: electron ionization cross section (source: lit. reference); $x_{\max}(\text{Exp})$: peak mole fraction measured in the experiment; $x_{\max}(\text{Model})$: simulated peak mole fraction; *h*: position of maximum in the experiment respectively simulations.

Species	M	E (eV)	IP (eV)	Calculated as	Calibration method	Ref.	$x_{\max}(\text{Exp.})$	<i>h</i> (mm)	$x_{\max}(\text{Model})$	<i>h</i> (mm)
CH ₂	14	12	10.396	Methylene radical	RICS (CH ₄)	[50]	5.05E-05	4.0	2.19E-04	9.0
CH ₃	15	10.6	9.84	Methyl radical	RICS (CH ₄)	[50]	2.61E-03	4.5	7.57E-03	5.9
CH ₄	16	13.4	12.61	Methane	Direct	[51]	1.57E-02	4.5	2.00E-02	5.2
C ₂ H ₂	26	10.6	11.4	Acetylen	Direct	[52]	2.12E-02	5.0	2.52E-02	6.9
C ₂ H ₃	27	10.6	8.59	Vinyl radical	RICS (C ₂ H ₄)	[53]	1.41E-05	2.5	–	–
C ₂ H ₄	28	12	10.5138	Ethylene	Direct	[54]	2.00E-02	4.0	2.16E-02	5.1
HCO	29	10.6	8.12	Formyl radical	Convolution	[53]	5.84E-05	4.0	1.88E-04	6.9
C ₂ H ₅	29	10.6	8.34	Ethyl radical	RICS (C ₂ H ₆)	[52]	1.01E-04	3.5	2.14E-04	4.9
CH ₂ O	30	12	10.88	Formaldehyde	Convolution	[53]	5.67E-03	3.5	6.50E-03	4.2
C ₂ H ₆	30	12	11.52	Ethane	Direct	[54]	3.15E-03	3.0	5.05E-03	4.3
CH ₃ O	31	12	10.72	Methoxy radical	RICS (CH ₃ OH)	[52]	7.07E-05	0.0	4.52E-05	2.8
CH ₃ OH	32	12	10.84	Methanol	Direct	[55]	7.28E-04	1.5	1.84E-04	3.1
C ₃ H ₂	38	12	10.43	Propadienylidene	Convolution	RICS (C ₃ H ₆)	7.70E-06	3.5	4.78E-04	6.7
C ₃ H ₃	39	10.6	8.68	Propargyl radical	RICS (C ₃ H ₆)	[53]	1.53E-04	4.5	2.71E-04	6.4
C ₃ H ₄	40	10.6	10.36	Propyne	RICS (C ₃ H ₆)	[53]	4.74E-03	4.5	2.86E-03	5.5
C ₃ H ₅	41	10.6	8.1	Allyl radical	RICS (C ₃ H ₆)	[53]	4.67E-04	4.0	3.91E-04	5.0
C ₂ H ₂ O	42	10.6	9.617	Ketene	Convolution	RICS (C ₂ H ₆ O)	2.10E-03	4.0	1.02E-03	4.8
C ₃ H ₆	42	10.6	9.73	Propene	Direct	[54]	4.47E-03	4.0	1.06E-02	4.6
C ₂ H ₄ O	44	12	10.229	Acetaldehyde	Convolution	RICS (C ₂ H ₆ O)	1.93E-03	3.5	2.59E-03	4.6
C ₃ H ₈	44	12	10.94	Propane	Direct	[52]	5.59E-04	2.5	3.79E-04	3.8
C ₂ H ₆ O	46	12	10.48	Ethanol	Direct	[55]	5.07E-04	1.0	8.51E-06	4.7
C ₄ H ₂	50	12	10.1	1,3-Butadiyne	Convolution	[53]	8.89E-04	5.5	1.27E-03	6.7
C ₄ H ₄	52	10.6	9.58	But-1-en-3-yne	Convolution	[53]	4.65E-04	4.5	6.19E-04	5.5
C ₄ H ₅	53	10.6	7.95	But-2-yn-1-yl radical	Convolution	RICS (1-butene)	6.47E-06	4.0	2.63E-05	5.4
C ₄ H ₆	54	10.6	9.072	1,3-Butadiene	Convolution	[53]	2.03E-03	4.0	1.30E-03	4.5
C ₃ H ₃ O	55	10.6	7	1-Oxoprop-2-enyl	RICS (C ₃ H ₆ O)	[52]	7.43E-06	3.5	1.38E-05	5.0
C ₄ H ₇	55	10.6	8.04	But-1-en-4-yl radical	RICS (n-C ₄ H ₈)	[52]	2.14E-04	3.0	6.94E-05	3.5
C ₃ H ₄ O	56	12	10.11	2-Propenal	RICS (C ₃ H ₆ O)	[52]	5.11E-04	2.5	4.61E-04	4.6
C ₄ H ₈	56	13.4	9.55	1-Butene	Direct	[52]	4.05E-04	3.0	1.17E-03	4.1
C ₃ H ₆ O	58	10.6	9.703	Acetone	Direct	[52]	1.55E-03	3.0	3.70E-04	4.1
C ₃ H ₈ O	60	12	10.22	1-Propanol	RICS (C ₃ H ₆ O)	[52]	1.53E-05	2.0	–	–
C ₅ H ₄	64	12	9.5	1,3-Pentadiyne	RICS (C ₅ H ₈)	[52]	1.52E-05	5.0	–	–
C ₅ H ₆	66	10.6	8.57	1,3-Cyclopentadiene	RICS (C ₅ H ₈)	[52]	5.41E-05	4.5	2.69E-05	4.3
C ₄ H ₄ O	66	12	8.88	Furan	RICS (2-MTHF)	[52]	1.07E-04	3.5	–	–
C ₅ H ₈	68	12	9.46	1,4-Pentadiene	Direct	[52]	1.56E-04	4.0	–	–
C ₄ H ₆ O	68	10.6	9.73	2-Butenal	RICS (2-MTHF)	[52]	2.72E-03	3.5	1.33E-03	4.4
C ₅ H ₁₀	70	10.6	9.04	2-Pentene	RICS (C ₅ H ₈)	[52]	4.22E-04	0.5	–	–
C ₄ H ₈ O	70	10.6	9.4	Tetrahydrofuran	RICS (2-MTHF)	[52]	8.99E-04	2.0	–	–
C ₆ H ₂	74	10.6	9.5	1,3,5-Hexatriyne	Convolution	[53]	2.34E-05	5.5	–	–
C ₆ H ₄	76	10.6	9.03	Benzene	Convolution	RICS (C ₆ H ₂)	7.18E-06	5.0	–	–
C ₆ H ₆	78	10.6	9.24	Benzene	Direct		3.79E-05	4.5	2.80E-05	5.8
C ₆ H ₈	80	10.6	8.25	1,3 Cyclohexadien	RICS (C ₆ H ₆)	[52]	1.04E-05	4.0	–	–
C ₆ H ₁₀	82	10.6	8.95	Cyclohexene	RICS (C ₆ H ₆)	[52]	4.49E-05	7.0	–	–
C ₅ H ₈ O	84	10.6	8.05	4,5-Dihydro-2-methylfuran	RICS (2-MTHF)	[52]	2.86E-04	2.0	2.41E-03	4.5
C ₇ H ₈	92	10.6	8.828	Toluene	Direct		1.38E-04	4.5	1.34E-06	6.1
C ₆ H ₆ O	94	10.6	8.49	Phenol	Convolution	RICS (Benzene)	3.33E-06	5.5	2.21E-06	5.8

More information on the included reactions and their rate coefficient expressions can be found in the mechanism files in the [Supplemental material](#). A list with the structures of relevant species, including their names in the mechanism and correct chemical nomenclature is also provided in the [Supplemental material](#) to clarify the naming of species.

4. Results and discussion

After a short discussion of major species and temperature profiles of the flames, a detailed analysis of the most important intermediate species will be given in the following section. An overview of species mole fractions and of modeling results is presented in [Table 3](#) for the EI-MBMS experiment and in [Table 4](#) for the PI-MBMS measurements, where peak mole fractions are reported together with their positions.

4.1. Major species

The major species, including 2-MTHF, O₂, H₂, CO, CO₂, and Ar are shown in [Fig. 2](#) for both experiments. For the simulations, the exhaust gas temperature was assumed to be close to the upper limit given by the experimental error; it was assumed to be 2160 K for the measured 2060 K (± 100 K) in the EI-MBMS experiment and taken as 1980 K for the measured 1880 K (± 100 K) in the PI-MBMS experiment. This choice while permitted within experimental uncertainty gave better agreement than that using the measured values. The resulting temperature profiles used for the simulation are included in [Fig. 2](#) (dashed lines) as well as in the [Supplemental material](#) together with the measured data. Also, equilibrium values calculated with the Gaseq program [44] are shown in [Fig. 2](#) at a

height of 33 mm. Experiment and simulation show reasonably good agreement. Discrepancies within the first five millimeters indicate faster fuel consumption in the experiment than predicted from the model. In this region some contributions from background gas in the ionization chamber cannot be excluded, in spite of careful corrections as discussed in Section 2. Regarding the paucity of data for this fuel and the estimates performed to establish the mechanism, the agreement in the general flame structure is a satisfactory basis for the interpretation of the fuel consumption and oxidation pathways.

4.2. Ring-opened intermediate species

As shown in [Fig. 1](#) and already described in Section 3, 2-MTHF can undergo several decomposition pathways. Under our conditions, about 70% of 2-MTHF destruction is initiated by abstraction of H-atoms by small radicals (H, O, OH, HO₂ etc.) that can form five different radicals (namely C₅H₉O-1, C₅H₉O-2, C₅H₉O-3, C₅H₉O-4, C₅H₉O-5) which can then undergo C–C-bond breaking by β -scission. Typically, the consumption of these first radicals is very fast so their concentrations are very low. Therefore, it is extremely difficult to measure these species (see also [25]). Additionally, ionization cross sections as well as ionization potentials are unknown for these different radicals, so that the quantification of them is currently not possible. As a consequence, detailed information about the occurrence of each radical structure cannot be successfully obtained even with the PI-MBMS experiment. Calculated reaction rates, shown in the reaction rate diagram in [Fig. 3](#), indicate that the dominant reaction (about 31%) in the first fuel consumption step is the reaction to C₅H₉O-5, where the radical is located and stabilized opposite the methyl group. The subsequent β -scission

Table 4

Isomer-resolved species measured with PI-MBMS in the 2-MTHF flame. *M*: nominal mass; *E*: photon energy; *IP*: ionization threshold; $\sigma(E)$: photoionization cross section (source: lit. reference, est. = estimated or * = unpublished work from Hansen et al.); $x_{\max}(\text{Exp.})$: peak mole fraction measured in the experiment; $x_{\max}(\text{Model})$: simulated peak mole fraction; *h*: position of maximum in the experiment respectively simulations.

Species	Formula	M	E (eV)	IP (eV)	σ	$x_{\max}(\text{Exp.})$	<i>h</i> (mm)	$x_{\max}(\text{Model})$	<i>h</i> (mm)
Methyl radical	CH ₃	15	9.96	9.84	[56]	6.09E–03	5.5	7.82E–03	5.6
Methane	CH ₄	16	13.16	12.61	[57]	8.79E–03	5.0	1.35E–02	4.9
Acetylene	C ₂ H ₂	26	11.46	11.4	[58]	2.50E–02	6.0	2.73E–02	6.3
Ethylene	C ₂ H ₄	28	10.96	10.5138	[58]	1.84E–02	5.0	1.86E–02	4.8
Allyl radical	C ₂ H ₅	29	8.96	8.117	[59]	6.86E–05	4.0	3.04E–04	4.6
Formaldehyde	CH ₂ O	30	10.96	10.88	[60]	4.51E–03	4.0	5.73E–03	4.0
Methanol	CH ₃ OH	32	10.96	10.84	[61]	5.22E–04	3.0	2.36E–04	3.1
Propargyl radical	C ₃ H ₃	39	8.96	8.67	[62]	3.08E–04	6.0	2.84E–04	6.1
Propyne	C ₃ H ₄	40	10.36	10.36	[58]	5.52E–04	5.0	1.11E–03	5.4
Allene	C ₃ H ₄	40	9.96	9.692	[58]	2.69E–04	5.5	1.14E–03	5.5
Propene	C ₃ H ₆	42	9.96	9.73	[29]	5.90E–03	4.5	1.04E–02	4.5
Ketene	C ₂ H ₂ O	42	9.66	9.617	*	1.65E–03	4.5	9.10E–04	4.6
Acetaldehyde	C ₂ H ₄ O	44	10.36	10.229	[58]	1.12E–03	4.5	1.83E–03	4.4
Ethenol	C ₂ H ₄ O	44	9.66	9.33	[58]	1.04E–04	4.0	1.13E–04	4.9
Diacetylene	C ₄ H ₂	50	10.36	10.17	[58]	3.43E–04	6.5	1.18E–03	6.0
Vinylacetylene	C ₄ H ₄	52	9.66	9.58	[58]	1.37E–04	6.0	5.32E–04	5.1
1,3-Butadiene	C ₄ H ₆	54	9.66	9.072	[58]	6.92E–04	5.0	1.24E–03	4.5
2-Propenal	C ₃ H ₄ O	56	10.36	10.11	Est.	3.28E–04	4.5	5.09E–04	4.2
1-Butene	C ₄ H ₈	56	9.66	9.55	[63]	2.73E–04	4.0	1.06E–03	4.0
<i>i</i> -Butene	C ₄ H ₈	56	9.66	9.22	[63]	4.93E–05	4.0	3.54E–05	4.4
2-Butene	C ₄ H ₈	56	9.66	9.09	[63]	3.93E–05	4.0	3.86E–04	3.3
<i>n</i> -Butane	C ₄ H ₁₀	58	10.96	10.57	[63]	3.13E–05	6.0	2.64E–05	3.8
Propanal	C ₃ H ₆ O	58	9.96	9.96	[63]	1.42E–03	4.5	4.58E–04	4.0
Acetone	C ₃ H ₆ O	58	9.96	9.703	[29]	1.85E–05	4.5	1.02E–04	4.6
1,3-Cyclopentadiene	C ₅ H ₆	66	8.96	8.57	*	3.13E–05	5.0	3.37E–05	4.2
1,4-Pentadiene	C ₅ H ₈	68	9.66	9.46	*	7.55E–05	3.0	–	–
Cyclopentene	C ₅ H ₈	68	9.66	9.01	[58]	2.52E–05	3.0	–	–
Furan	C ₄ H ₄ O	68	9.66	8.82	*	2.23E–05	3.0	–	–
2-Butenal	C ₄ H ₆ O	70	9.96	9.73	*	1.79E–03	1.5	8.25E–04	4.9
2,5-Dihydrofuran	C ₄ H ₆ O	70	9.66	9.1	*	1.14E–04	1.0	6.19E–06	4.2
2,3-Dihydrofuran	C ₄ H ₆ O	70	8.96	8.3	*	5.89E–05	3.5	7.02E–04	4.0
Tetrahydrofuran	C ₄ H ₈ O	72	9.66	9.4	[63]	6.25E–04	1.0	–	–
Benzene	C ₆ H ₆	78	9.66	9.24	[58]	2.45E–05	4.0	1.94E–05	5.2

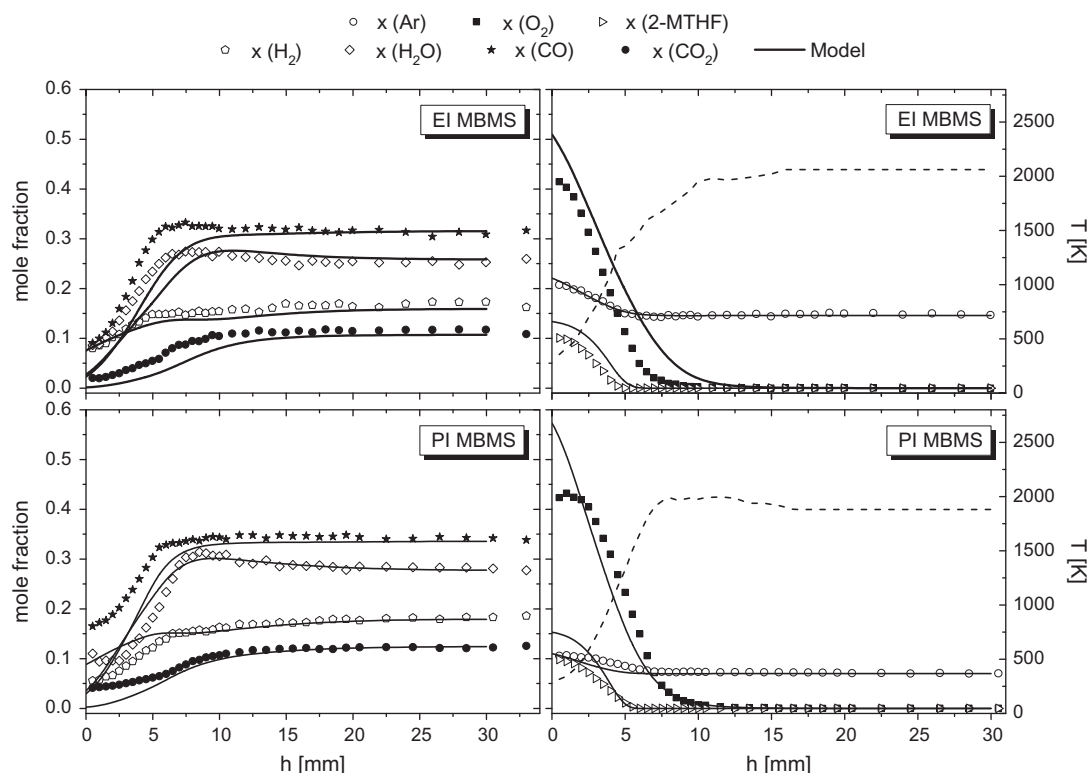


Fig. 2. Main species mole fraction and perturbed temperature (T) profiles for the 2-MTHF flames of the EI-MBMS and PI-MBMS experiments as a function of height (h). Symbols represent the experimental data, lines the modeling results. Equilibrium values are indicated at $h = 33$ mm. Dashed lines correspond to temperature profiles.

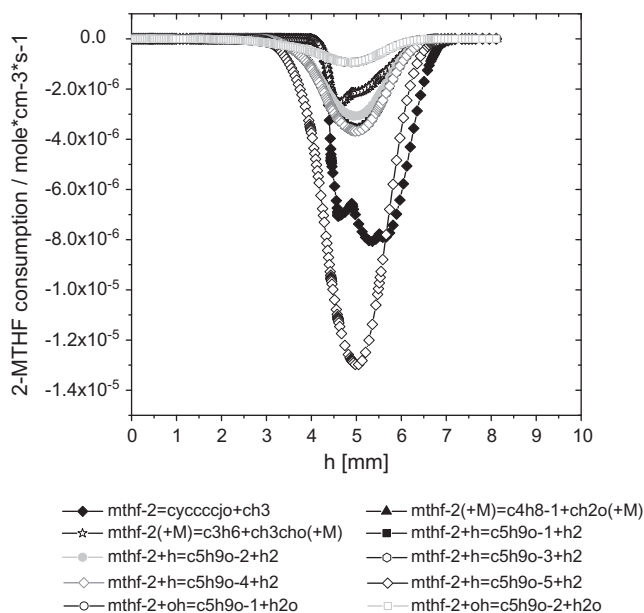


Fig. 3. Reaction rate diagram of the first fuel consumption step.

leads to ring opening and results in the direct production of propene (C_3H_6) and the CH_2CHO radical which can stabilize to acetaldehyde (CH_3CHO) by H-atom addition. The same species will be formed by different ring-opening reactions from the $\text{C}_5\text{H}_9\text{O-4}$ radical. Formation of propene then results from H-atom addition to the allyl radical (C_3H_5) which can also stabilize to allene ($\text{a-C}_3\text{H}_4$). In analogy to tetrahydrofuran and the three reaction routes reported by Dagaut et al. [45] for its decomposition in shock tubes,

similar reaction pathways can be assumed for 2-MTHF. As seen from the reaction scheme in Fig. 1, two different bimolecular decomposition reactions are possible and contribute with 5% each to the first breakdown step of 2-MTHF. One of them results also in the formation of propene and acetaldehyde.

Figure 4 therefore presents mole fraction profiles for C_3 -compounds including C_3H_4 , C_3H_5 , and C_3H_6 as well as for acetaldehyde that would be formed in conjunction. The C_3H_6 profile shows an acceptable agreement between experiment and model; the model slightly over-predicts mole fractions of both experiments but is within the stated error range. The quantitative model prediction of the allyl radical, which was only quantified in the EI-MBMS experiment, is in very good agreement and the profile shape is quite well captured. Both predicted peak mole fractions show again a shift of about 1.5 mm against experimental data of the EI-MBMS experiment as it could be already seen for the major species. The C_3H_4 signal includes contributions from allene ($\text{CH}_2=\text{C}=\text{CH}_2$) and propyne ($\text{CH}\equiv\text{CH}-\text{CH}_3$) which cannot be separated in the EI-MBMS experiment. The EI-MBMS evaluation used the propyne cross section because isomer separation with PI-MBMS resulted in higher mole fractions of propyne. The sum of the mole fractions of both isomers in model and experiment is in acceptable agreement. The mechanism predicts similar peak mole fractions for both isomers but slightly different profiles. On closer inspection, the shape of the mole fraction profiles is reasonably well matched also in the experimental data, but they show different behavior in their maximum concentrations. Propyne exists about a factor of 2.5 more than allene, and while the model prediction of propyne is just in an acceptable agreement, allene is over-predicted.

The signal of $\text{C}_2\text{H}_4\text{O}$ can be related to acetaldehyde (IP: 10.229 eV) as well as to ethenol (IP: 9.33 eV) and ethylene oxide (IP: 10.56 eV). As can be seen from the photoionization energy (PIE) curve in Fig. 4, which was measured at $h = 5$ mm in the flame,

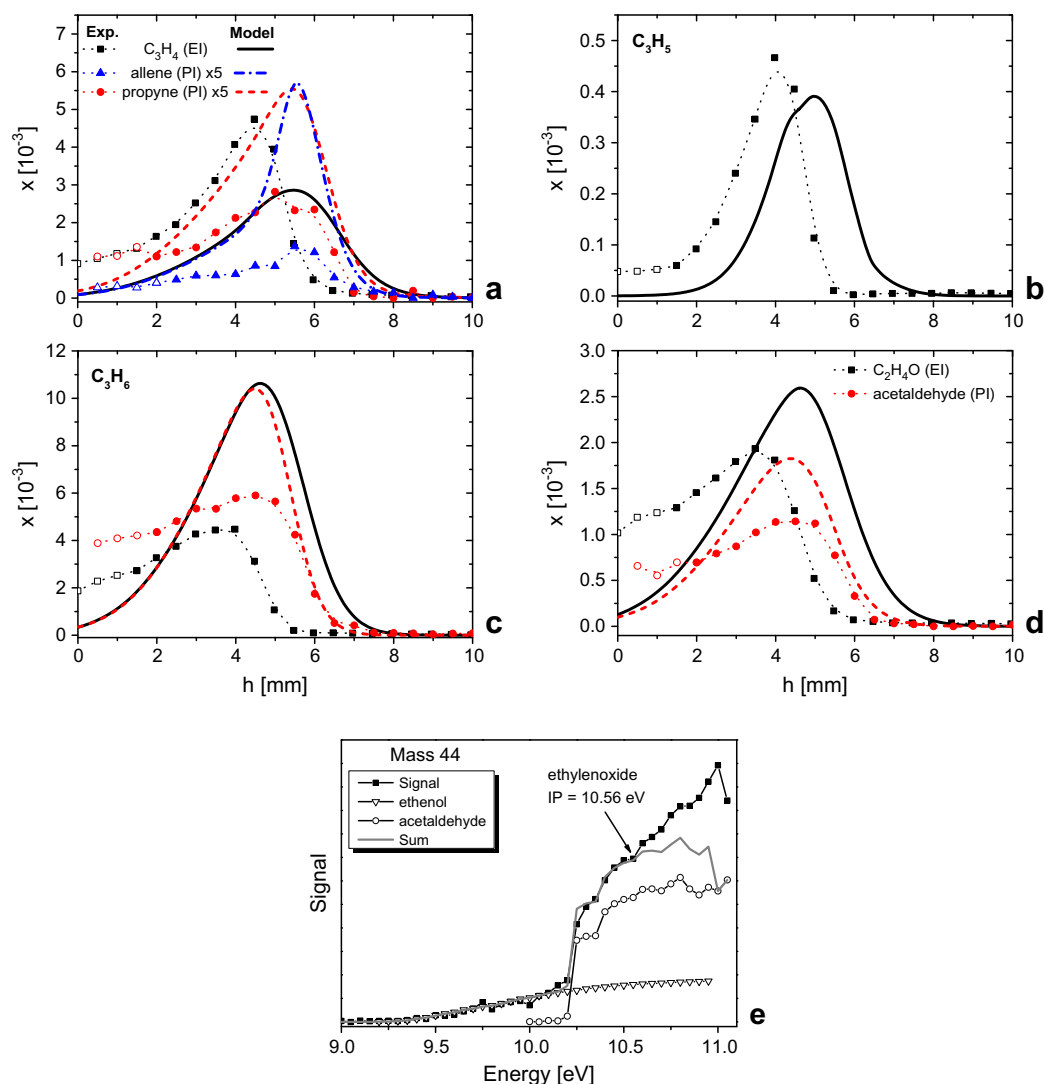


Fig. 4. Mole fraction profiles for selected C_3 -species (a–c) and C_2H_4O (d). Symbols and thin dotted lines represent experimental results, thick lines the corresponding simulations. Measurements at $h < 1$ mm (EI-MBMS), respectively $h < 1.5$ mm (PI-MBMS) labeled by open symbols may be affected by sampling probe effects. Note that model and experiment for allene and propyne are scaled by a factor of 5. Flame-sampled PIE curve of $m/z = 44$ (e). Closed symbols (connected to guide the eye) represent the measured signal, open symbols show the scaled photoionization cross section of the respective species. The obtained sum signal is shown as continuous grey line.

all three species contribute to the measured signal of mass 44 and they can be identified clearly in the PI-MBMS experiment by their ionization potentials. Acetaldehyde contributes to the signal about 20 times more than ethenol. Unfortunately, with cross sections unavailable for ethylene oxide, this species cannot be quantified. Nevertheless, with acetaldehyde as the dominant species, a direct comparison of the EI-MBMS result for C_2H_4O to acetaldehyde in the model is reasonable, too. In both cases the model is in good agreement with the experimental data. The difference between the experimental mole fraction profiles due to the difference in the flame conditions is well reflected in the model (maximum mole fractions of experiments differ by a factor of 1.7 and a factor of 1.4 in the simulations). Again, the shift to larger heights of the simulation versus the experimental data is more pronounced in the EI-MBMS case.

The decomposition via H-atom abstraction and subsequent β -scissions of the radicals C_5H_9O -1 and C_5H_9O -3 which contribute each 10% to the decomposition of 2-MTHF leads to the formation of 1-butene ($CH_3-CH_2-CH=CH_2$), 1,3-butadiene ($CH_2=CH-CH=CH_2$), and 2-butene ($CH_3-CH=CH-CH_3$) via the stabilization of the C_4H_7 radical. Additionally, formaldehyde

(CH_2O) is a product during the formation of the C_4H_7 radical. As already described above, two possible pathways exist of thermally-induced bimolecular decomposition, one of which results in propene and acetaldehyde (see above), the other one is responsible for the formation of formaldehyde and 1-butene. Fig. 5 shows these C_4 -species together with formaldehyde. For the C_4H_7 radical, the mole fraction profile from the EI-MBMS experiment includes several possible isomers, namely $CH_2=CH-CH_2-\dot{C}H_2$ (name in the mechanism: c4h71-4), $CH_2=CH-\dot{C}H-CH_3$ (c4h71-3), $CH_3-CH=CH-\dot{C}H_2$ (c4h72-1), $CH_3-CH_2-\dot{C}H=CH$ (c4h71-1), $CH_3-CH_2-\dot{C}=CH$ (c4h71-2), and $CH_3-C(CH_2)-\dot{C}H_2$ (ic4h7), of which the first three isomers can be formed through subsequent different β -scissions of ring-opened C_5H_9O radicals (created by ring-opening of C_5H_9O -1 and C_5H_9O -3, see Fig. 1). Regarding the sum signal of C_4H_7 radical in Fig. 5, it is notable that the simulated mole fraction profiles show a contrasting behavior to the experiment around 5 mm. To understand the occurrence of this decelerated decomposition of the C_4H_7 radical, Fig. 5 also includes isomer-resolved simulated mole fraction profiles of the C_4H_7 radicals considered by the mechanism. Photoionization cross sections to separate these isomers experimentally are, however, not

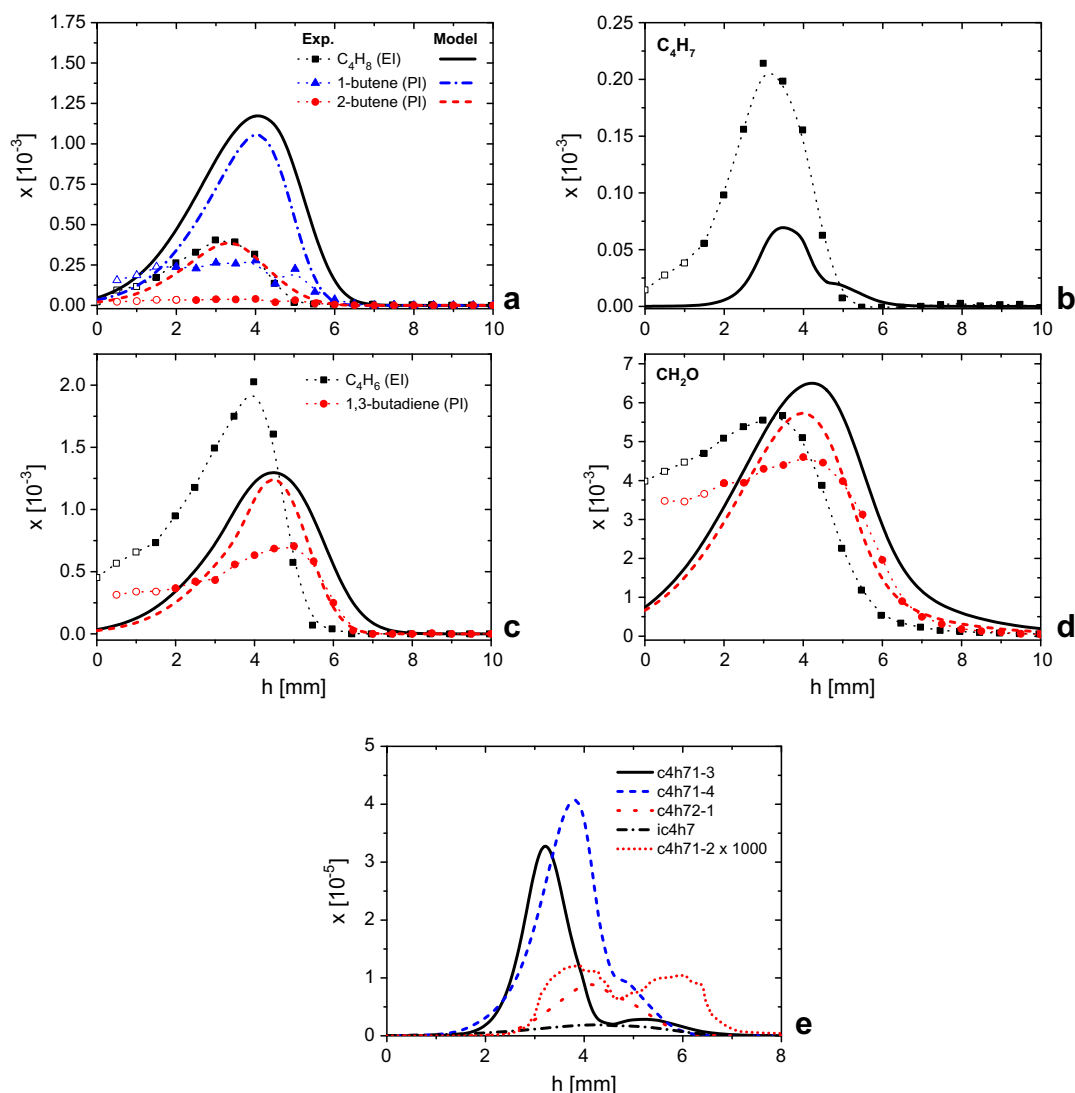


Fig. 5. Mole fraction profiles for selected C_4 -species (a–c) and CH_2O (d). Symbols and thin dotted lines represent experimental results, thick lines the corresponding simulations. Measurements at $h < 1$ mm (EI-MBMS), respectively $h < 1.5$ mm (PI-MBMS) labeled by open symbols may be affected by sampling probe effects. For C_4H_7 experimental data from the PI-MBMS experiment are not available. Note that experimental results of 2-butene are scaled by a factor of 10. Isomer selective simulated mole fraction profiles of C_4H_7 radicals (e).

available. The simulations show that each isomer reaches its peak mole fraction at a different height. The isomers which are formed through ring-opening and subsequent β -scission (c4h71-4, c4h71-3, and c4h72-1) reach their maximum mole fractions between 3 and 4 mm, approximately coincident with the experimental peak position at 3 mm. However, c4h71-3 and c4h71-4, which are the largest contributions to the sum signal, show an increased formation, respectively a decelerated decomposition, around 5 mm. At this position the temperature of the flame is about 1350 K and H-atom abstraction by H, O, CH_3 , CH_3O radicals etc. (for details see mechanism in the [Supplemental material](#)) from 1-butene (c4h8-1) which is also produced directly through thermal decomposition of the fuel molecule becomes important, forming c4h71-3 and c4h71-4. Below this temperature, C_4H_7 radical formation is predominantly controlled by ring-opening and β -scission. In addition, ic4h7 contributes to the simulated sum signal but is less sufficient to explain the difference in the profile shape between experiment and simulation. The resulting shoulder in the simulated profile for these contributions to the sum of C_4H_7 is not evident in the EI-MBMS data. This difference in peak values may have its cause in the necessary choice of a single isomer cross section for

the experimental data evaluation, in this case based on the assumption that C_4H_7 is but-1-en-4-yl radical (c4h71-4).

Besides the thermally induced reaction to 1-butene via thermal decomposition of 2-MTHF there are two other possible pathways for the formation of 1-butene by stabilization of the C_4H_7 radical (via C_5H_9O -1 and C_5H_9O -3, see [Fig. 1](#)). In contrast, there is only one for the formation of 2-butene (via C_5H_9O -3), so that the production of 1-butene against 2-butene should be preferred. The model predicts a mole fraction of 1-butene which is about six times higher than that of 2-butene (EI-MBMS conditions), respectively three times higher for the PI-MBMS conditions, potentially caused by the different temperature profiles of the flames. As determined individually in the PI-MBMS experiment, 1-butene is the more prominent species, with a mole fraction ratio over 2-butene of about a factor of 7. Note that mole fractions from EI- and PI-MBMS for C_4H_8 are of the same order of magnitude whereas the agreement between the model and experiment is less satisfactory in this case. However, also the simulations agree in their maximum values.

The different C–C bond scission of ring-opened C_5H_9O radicals which form the described C_4H_7 radicals lead to the formation of

formaldehyde at the same time. The quantitative agreement between the experimental results, which includes also the formation of CH_2O by bimolecular decomposition, and the model prediction is remarkable. A tendency is seen in both experiment and simulation that the peak mole fraction of C_4H_7 and CH_2O occur at lower heights than that of C_4H_6 , in accord with the subsequent formation of C_4H_6 in the combustion process.

The largest contribution to the C_4H_6 signal is from 1,3-butadiene as identified in the PI-MBMS experiment. The energy resolution was not sufficient to distinguish between 1,3-butadiene and 1,2-butadiene, but we chose to evaluate this signal as 1,3-butadiene. Contributions from 1-butyne to C_4H_6 were separated in the PI-MBMS experiment (from a burner scan at different energy); also, contributions from 2-butyne were taken into account using the PIE spectrum. Figure 5 provides the 1,3-butadiene mole fraction profile for the PI-MBMS experiment in comparison with the simulation, which is in good agreement. Again, isomer separation is not possible in the EI-MBMS experiment. Mole fractions from this experiment were evaluated as 1,3-butadiene and are slightly higher than the simulations, potential contributions from 1-butyne

and 2-butyne cannot be taken into account. Overall, agreement between experiments and simulations is satisfactory and within the error limits.

The ring opening of $\text{C}_5\text{H}_9\text{O}-2$ and the subsequent β -scission of the radical leads to the formation of ethylene (C_2H_4) and the 2-oxopropyl-radical (CH_3COCH_2). The latter can stabilize to acetone (CH_3COCH_3) via H-atom addition or decomposes to ketene ($\text{CH}_2=\text{C}=\text{O}$) and the CH_3 radical. Note that fuel consumption via the reaction pathway over $\text{C}_5\text{H}_9\text{O}-2$ is the only one which forms directly the CH_3COCH_2 radical whereas ethylene is formed also from different pathways. These additional routes include the 2-MTHF decomposition via $\text{C}_5\text{H}_9\text{O}-1$ as well as the cleavage of the methyl group of the fuel and the subsequent ring-opening and β -scission (see below). Relevant C_2 -species (namely: acetylene, ethylene and ketene) as well as $\text{C}_3\text{H}_6\text{O}$ species are shown in Fig. 6. Both, experiment (PI-MBMS) and model predict a higher peak mole fraction for propanal ($\text{CH}_3-\text{CH}_2-\text{CH}=\text{O}$) than for acetone. Nevertheless, acetone is over-predicted by the simulation whereas the predicted mole fractions for propanal are smaller than the experimental results. Experimental separation of these two isomers was

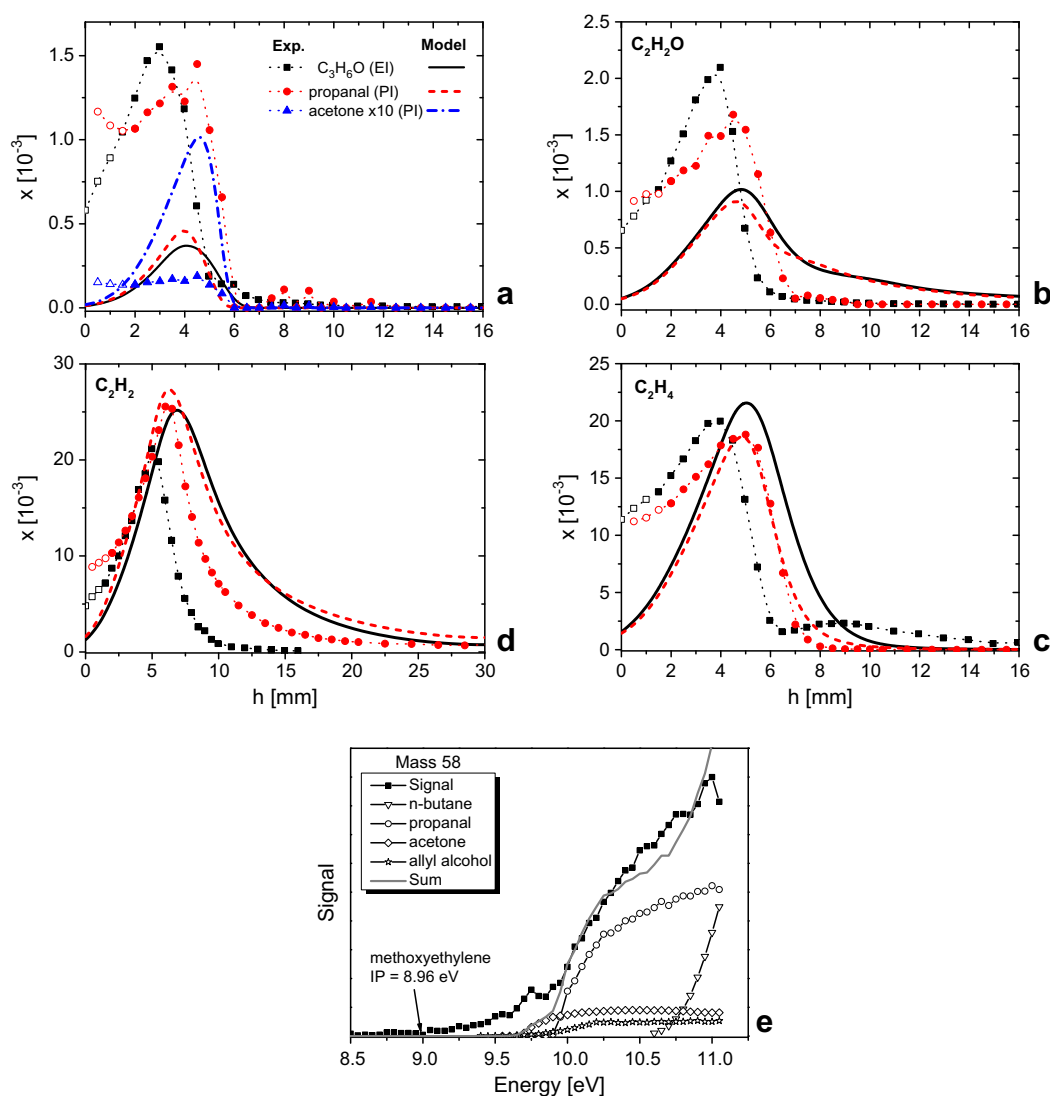


Fig. 6. Mole fraction profiles of $\text{C}_3\text{H}_6\text{O}$ (a) and of selected C_2 -species (b–d). Symbols and thin dotted lines represent experimental results, thick lines the corresponding simulations. Measurements at $h < 1$ mm (EI-MBMS), respectively $h < 1.5$ mm (PI-MBMS) labeled by open symbols may be affected by sampling probe effects. Note, that model and experimental results of acetone from the PI-MBMS experiment are scaled by a factor of 10. Flame-sampled PIE curve of $m/z = 58$ (e). Closed symbols (connected to guide the eye) represent the measured signal, open symbols show the scaled photoionization cross section of the respective species. The obtained sum signal is shown as continuous grey line.

difficult here because their ionization potentials (acetone: 9.703 eV, propanal: 9.96 eV) are close and the data evaluation had to be based on the same measurement with 9.96 eV as fixed ionization energy. The relevant contributions of these species to the signal had to be evaluated from the PIE spectrum at $m/z = 58$, also presented in Fig. 6. Within the limited mass resolution of the PI-MBMS instrument, several additional species contribute to the total signal. The representation of the measured signal by weighted contributions from these species thus depends on numerous parameters. Reasonable agreement is reached when the propanal contribution is a factor of five higher than that of acetone, with remaining discrepancies in the range of 9.0–9.8 eV, however. Unfortunately, for methoxyethylene ($\text{CH}_2=\text{CH}-\text{O}-\text{CH}_3$) which has an IP of 8.96 eV, photoionization cross section data are not available, and improvement of the fit near 9.0 eV is thus presently not possible. In the EI-MBMS experiment, *n*-butane was separated from the oxygenated species by mass. The predicted sum of acetone and propanal is a factor of 4 lower than the EI-MBMS result for $\text{C}_3\text{H}_6\text{O}$ but still within the specified error range. The quantitative model prediction of the other destruction products shown in Fig. 6 is reasonable. Profile shapes for ketene, ethylene, and acetylene are in reasonable agreement with the experimental results when the first data points close to the burner are not considered because of probe effects. Again, a shift of modeled species profiles to higher heights is observed in comparison to the EI-MBMS data.

4.3. Non-ring-opened structures

As already described in Section 3, the produced fuel radicals with the radical site on one of the ring carbon atoms can react with available radicals to one of the three 2-methyl-dihydrofuran isomers, namely 2,3-dihydro-2-methyl-furan ($\text{cyc}^*\text{ccc}(\text{c})\text{o}$), see list of structures in the Supplemental material), 2,3-dihydro-5-methyl-furan ($\text{cyccc}^*\text{c}(\text{c})\text{o}$), and 2,5-dihydro-2-methyl-furan ($\text{cycc}^*\text{cc}(\text{c})\text{o}$). Particularly $\text{C}_5\text{H}_9\text{O}-3$ and $\text{C}_5\text{H}_9\text{O}-4$ which are non-resonantly stabilized radicals tend to stabilize easier to these non-ring-opened species. Separation and quantification of the named isomers by the PI-MBMS experiment fails in this case due to lack of known ionization potentials as well as photoionization cross sections. $\text{C}_5\text{H}_8\text{O}$ signals from EI-MBMS were evaluated as 2,3-dihydro-5-methyl-furan and its IP was adapted from [25]. The resulting mole fraction profile is presented in Fig. 7 together with that of $\text{C}_4\text{H}_6\text{O}$, to which 2,3-dihydrofuran, 2,5-dihydrofuran and 2-butenal (crotonaldehyde) can contribute. Experimental $\text{C}_5\text{H}_8\text{O}$ results are over-predicted in the simulations, which could be explained by the assumption of pure 2,3-dihydro-5-methyl-furan in the data evaluation. As already mentioned, particularly 2,3-dihydrofuran is an important combus-

tion intermediate for 2-MTHF as it can be produced by CH_3 elimination from the primary radical $\text{C}_5\text{H}_9\text{O}-3$ or by elimination of the CH_3 group from the fuel and subsequent H-atom abstraction. The loss of the CH_3 group from the fuel forming $\text{C}_4\text{H}_7\text{O}$ radical is the second most important pathway of the decomposition of 2-MTHF and contributes to 18% of it under our conditions. However, subsequent ring opening seems to be faster than the abstraction of the nearby H-atom and yields predominantly ethylene and CH_2CHO radical (by C–O bond breaking) or ethylene and ketene (by C–C bond breaking). As can be expected from the reaction scheme in Fig. 1, the formation of 2,3-dihydrofuran should be preferred over 2,5-dihydrofuran regarding the position of the radical in the precursor radical. As already mentioned above, Lifshitz et al. have presented the isomerization of 2,3-dihydrofuran to cyclopropane carboxaldehyde and further to 2-butenal (crotonaldehyde) [14]. In contrast, the 2,5-dihydrofuran did not isomerize under similar conditions in their shock-tube experiments. The isomerization reaction leading to 2-butenal was also observed in premixed tetrahydrofuran flames [25]. Also, the flame-sampled PIE curve of $m/z = 70$ presented in Fig. 7 confirms that 2-butenal is formed. The isomer distribution in Fig. 7 considers the PI cross section curves of 2,3-dihydrofuran, 2,5-dihydrofuran, 2-butenal, and 1-pentene which are fitted to the observed signal. Good agreement with the measured signal was achieved when a 3.5 times higher mole fraction of 2,5-dihydrofuran than of 2,3-dihydrofuran was assumed. The initially unexpected higher mole fraction of 2,5-dihydrofuran suggests that most of the 2,3-dihydrofuran might isomerize to 2-butenal, an observation that is confirmed by the dominant mole fraction of 2-butenal (see Table 4). Therefore the EI-MBMS data were evaluated as 2-butenal, which might lead to a slight under-prediction of the obtained mole fraction for $\text{C}_4\text{H}_6\text{O}$ by the model.

The same initial fuel decomposition reaction (loss of the CH_3 group from the fuel) can lead to the formation of THF by abstraction of an H-atom from any other present species to the non-ring-opened $\text{C}_4\text{H}_7\text{O}$ radical. Besides THF, a number of other species could contribute to the $\text{C}_4\text{H}_8\text{O}$ signal of the EI-MBMS experiment. The PIE curve for $m/z = 72$ in Fig. 8 shows that the measured signal from the corresponding PI-MBMS experiment is excellently fitted at energies below 9.8 eV using only the cross section of THF. The observed deviation near 9.8 eV coincides with the ionization threshold of *n*-butanal (IP: 9.82 eV) that can be produced through the same reaction channel by H-atom addition to the ring-opened $\text{C}_4\text{H}_7\text{O}$ radical. Most other conceivable species have ionization potentials below this value. It seems thus obvious from the PI-MBMS experiment that THF and *n*-butanal are the predominant species. In the absence of photoionization cross sections for *n*-butanal, the measured PIE curve cannot be fitted above 9.8 eV. The

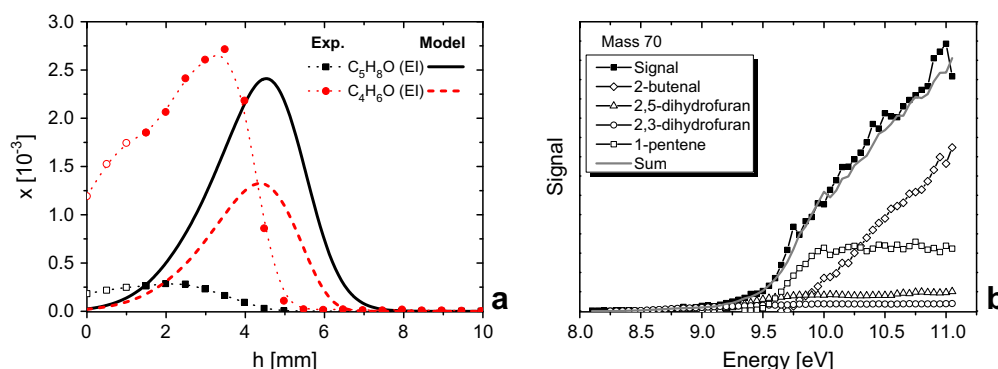


Fig. 7. Mole fraction profiles of $\text{C}_4\text{H}_6\text{O}$ and $\text{C}_5\text{H}_8\text{O}$ obtained from EI-MBMS (a). Symbols and thin dotted lines represent experimental results, thick lines the corresponding simulations. Measurements at $h < 1$ mm (EI-MBMS) labeled by open symbols may be affected by sampling probe effects. Flame-sampled PIE curve of $m/z = 70$ (b). Closed symbols (connected to guide the eye) represent the measured signal, open symbols show the scaled photoionization cross section of the respective species. The obtained sum signal is shown as continuous grey line.

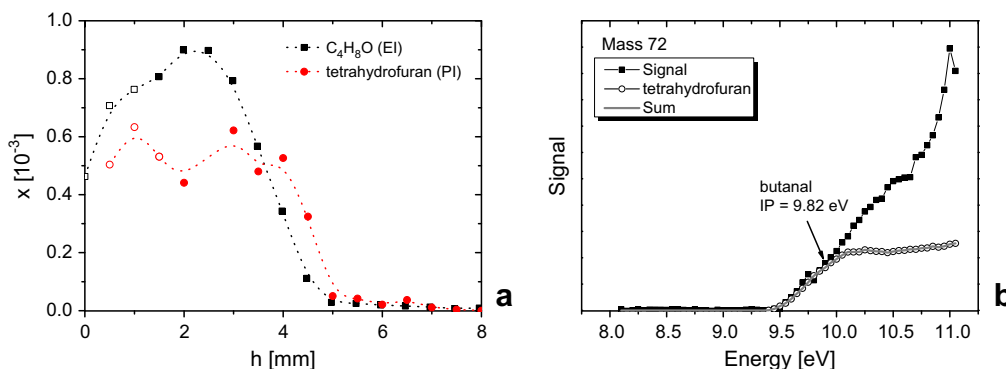


Fig. 8. Mole fraction profiles of C_4H_8O (sum of isomers) obtained from EI-MBMS and of tetrahydrofuran measured by PI-MBMS (a). Measurements at $h < 1$ mm (EI-MBMS), respectively $h < 1.5$ mm (PI-MBMS) labeled by open symbols may be affected by sampling probe effects. Flame-sampled PIE curve of $m/z = 72$ (b). Closed symbols (connected to guide the eye) represent the measured signal, open symbols show the scaled photoionization cross section of tetrahydrofuran. The obtained sum signal is shown as continuous grey line.

resulting mole fraction profiles of THF (from PI-MBMS) and C_4H_8O (from EI-MBMS, evaluated as THF) are presented in Fig. 8. Although several reactions for the formation of THF are included in the mechanism (for details see Supplemental material) the model fails in this case, predicting mole fractions below the absolute set threshold of 10^{-9} . However, the included reactions don't have any influence on simulations as radical decompositions are much faster in the temperature range where they become important.

In the discussion of potential pollutants resulting from fuel destruction and oxidation steps, carbonyl compounds are especially interesting, since they are known to impact air quality and photochemical reactions in the atmosphere [46]. In recent studies of automobile exhaust, Fontaras et al. [47] have detailed the potential influence of tailpipe emissions resulting from biodiesel addition to conventional diesel fuel and have detected about 12 carbonyl compounds, some of which have also been noted in the present study. Especially 2-butenal and acetaldehyde have appreciable mole fractions (based on the EI-MBMS results) of 2.72×10^{-3} and 1.93×10^{-3} , respectively. Mole fractions of acetaldehyde are about one order of magnitude larger than that in the non-oxygenated *n*-butane (3.9×10^{-4}) and *iso*-butane (2.7×10^{-4}) flames [27] but below those observed in 1-butanol (1.7×10^{-2}), 2-butanol (7.8×10^{-3}) and *iso*-butanol (3.9×10^{-3}) flames [26] which were investigated under similar conditions.

4.4. Build-up species and soot precursors

While the species discussed above can be formed by direct reactions coming from the fuel molecule with subsequent β -scissions and stabilization reactions via H-atom abstraction or addition, non-oxygenated C_5 - and C_6 -species must be formed by build-up reactions. Of this category, especially 1,3-cyclopentadiene and benzene are of interest because of their role as potential soot precursors; similarly, phenol may be important in this context. Figure 9 therefore presents the mole fraction profiles of the isomer sums of C_5H_6 , C_6H_6 and C_6H_6O . Since the detection of higher masses is more sensitive with the EI-MBMS system, only isomer mole fraction sums are given here; these were assumed to be 1,3-cyclopentadiene (C_5H_6), benzene (C_6H_6), and phenol (C_6H_6O) in the quantitative evaluation. The presence of each of these species was verified from PIE spectra at the corresponding mass ($m/z = 66, 78, 94$) with the PI-MBMS system. As already mentioned in Section 3, a subset to describe the formation of such aromatic species was included in the mechanism, which considers their production and consumption reactions (for details see Supplemental material). The predicted mole fractions and the profile shapes of

these aromatic species are in good agreement with the experimental data. Again the already discussed shift of the maxima to higher heights is observed, and it is more pronounced for benzene and phenol than for 1,3-cyclopentadiene.

Since the combination of propargyl radicals (C_3H_3) is an effective channel for benzene formation [48], C_3H_3 mole fraction profiles are included in Fig. 9. For the PI-MBMS setup, the predicted mole fractions as well as the shape are in excellent agreement with the experimental data. In the EI-MBMS experiment where C_3H_3 was calibrated with the RICS [32] approach from propene, mole fractions are slightly over-predicted by the model. Similar trends for C_3H_3 between model and experiment were already observed before [27].

Hansen et al. [49] discussed different benzene formation pathways for cyclic and non-cyclic hydrocarbon fuels and gave an upper limit for benzene that can be formed through $C_3 + C_3$ reactions. For this estimation they correlated the maximum mole fraction of benzene to that of the propargyl radical. For a given C_3H_3 mole fraction, they observed a higher benzene formation for fuels containing a ring structure (in their study: cyclohexane and methylcyclopentane) and reasoned that direct routes via dehydrogenation and ring-enlargement reactions are also responsible for the production of benzene. However, non-cyclic fuels showed an upper limit of benzene formation which correlated linearly with the observed maximum mole fraction of C_3H_3 . For 2-MTHF as a heterocyclic fuel, we found it interesting to see its behavior in this context. The experimental results as well as the correlated mole fractions of the simulated results here seem to be in agreement with the determined limit of benzene formation for non-cyclic fuels, suggesting that the formation of benzene in the combustion of 2-MTHF should predominantly involve $C_3 + C_3$ reactions.

Recently, the combustion of furan and its derivatives 2-MF and 2,5-DMF were investigated [11,13,19] under premixed low-pressure flame conditions using the same analytical technique (MBMS) and equipment (EI-MBMS experiment) as in the present study. While the chosen stoichiometry of their flames was the same as in our study, their dilution was 50% argon. In [19] a comparison of these fuels with respect to their pollutant potential is reported. Compared to 2-MTHF, the three furanic fuels show much higher mole fractions of benzene (furan: 1.3×10^{-4} ; 2-MF: 2.8×10^{-4} ; 2,5-DMF: 1×10^{-3}) than the maximum of 3.79×10^{-5} seen here, especially considering their higher argon dilution. Compared to these unsaturated cyclic fuels, combustion of the saturated 2-MTHF fuel suggests a lesser propensity of aromatics and soot formation. This trend is similar for the THF flame of Kasper et al. [25]. For their flame with $\phi = 1.75$, the maximum benzene mole

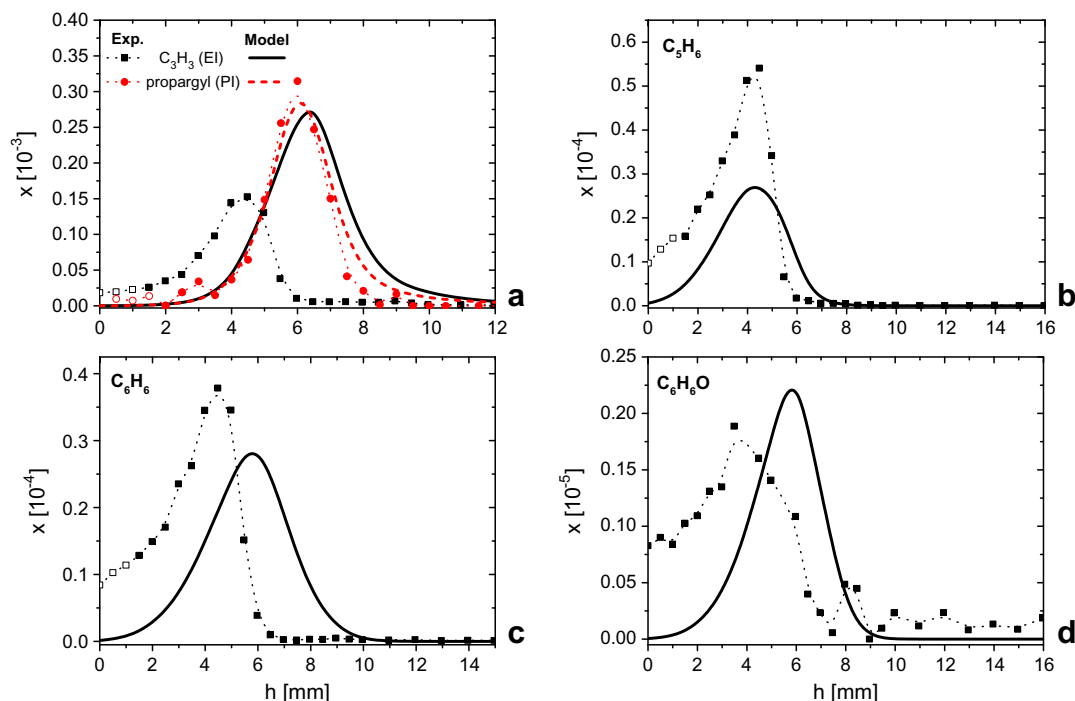


Fig. 9. Mole fraction profiles for the C₃H₃ radical (a) and selected aromatic species (b–d). Symbols and thin dotted lines represent experimental results, thick lines the corresponding simulations. Measurements at $h < 1$ mm (EI-MBMS), respectively $h < 1.5$ mm (PI-MBMS) labeled by open symbols may be affected by sampling probe effects. Experimental data of the cyclic species are only available from EI-MBMS measurements.

fraction is $\sim 1 \times 10^{-4}$, about two times higher than in the 2-MTHF flame here, but still below all measured benzene mole fractions detected in flames of the unsaturated furanic fuels.

Several other C₅- and C₆ build-up species were detected and quantified in the experiment, including e.g. C₅H₄, C₅H₈, C₅H₁₀, C₆H₂, C₆H₄, C₆H₈, C₆H₁₀. Their mole fractions as well as the position of their peak maximum are listed in Table 3 (EI-MBMS experiment) and Table 4 (PI-MBMS experiment). These species were not included in the present model.

5. Summary and conclusions

Recent interest in the combustion of cyclic ethers has demonstrated the need for the development of chemical kinetic models. Particularly, furan and its derivatives have recently received increased attention because they can be synthesized from biomass-derived carbohydrates [2] as potential transportation fuels or fuel additives. We have therefore performed a detailed study of low-pressure, premixed laminar 2-methyltetrahydrofuran (2-MTHF)/oxygen/argon flames with an equivalence ratio of 1.7 using molecular-beam mass spectrometry. The combination of a high mass resolution (EI-MBMS) and a synchrotron-based high energy resolution (PI-MBMS) setup enabled the quantification and identification of about 50 species as a function of burner distance, including radical and isomeric species. Including several sub-mechanisms [35–40] the obtained results were used for the development of a kinetic model for the high-temperature combustion of 2-MTHF. The entire mechanism consists of 185 species and 1412 reactions, with 25 species and 168 reactions added for the 2-MTHF sub-mechanism. To compare the predicted mole fractions with the experimental results, perturbed temperature profiles were used as input parameter. Good general agreement of the model with the experiments was achieved for the major species as well as for many combustion intermediate species. This good agreement is valid for profile shapes as well as for the quantitative mole fraction results within

experimental uncertainty. For PI-MBMS data the position of the peak maximum was excellently predicted for most intermediates whereas for EI-MBMS results a shift of up to 2 mm was observed in most cases. Regarding potentially harmful emissions from combustion of this fuel, several aldehydes and soot precursors were noted. Especially the formation of benzene as the first aromatic species seems to be lower than from the corresponding unsaturated ether fuels, namely furan and its derivatives, under similar conditions, suggesting superior performance of 2-MTHF in this respect. While the mechanism represents many features of the pre-mixed flame structure and combustion pathways adequately, some adjustments may be needed to include, for example, the formation of tetrahydrofuran which was clearly identified by the experiment but could not be predicted in adequate amounts by the mechanism.

Acknowledgments

K.M. and K.K.H. acknowledge the opportunity to perform the PI-MBMS measurements at the Advanced Light Source at Berkeley as part of the “Flame Team”, headed by Dr. Nils Hansen, Sandia National Laboratories, Livermore, USA. These authors wish to thank him for his continuing support and helpful discussion. The Advanced Light Source is supported by the Director, Office of Science, Office of Basic Energy Sciences, of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231. The work at Aachen is part of the Cluster of Excellence “Tailor-Made Fuels from Biomass”, which is funded by the Excellence Initiative by the German federal and state governments to promote science and research at German universities.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.07.006>.

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